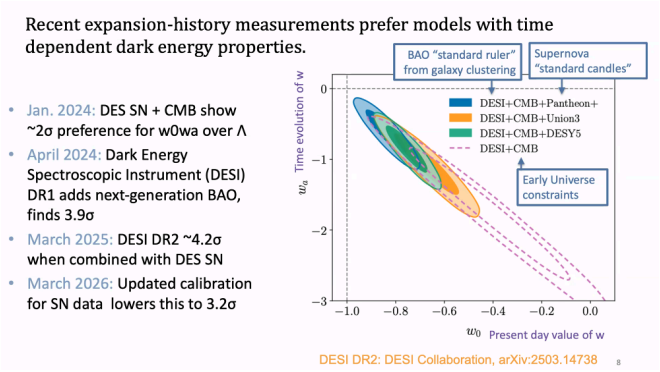
Professor Jessica Muir Colloquium Notes - 11/10/2026

Lambda CDM: A cosmological standard model in which lambda is the cosmological constant dark energy, and DM is cold dark matter. They have modeled dark energy and dark matter to be 68% and 27% of the universe, while the remaining 5% would be the visible matter. Our accepted best model of things, however, has open questions like is dark energy a cosmological constant or does general relativity need to be extended for dark energy.

Conclusions and Outline: it is an exciting time to study dark energy cosmology because of new observing efforts accompanied by recently found results that will be discussed below. Expansion history probes of the university hint at time-dependent properties over a cosmological constant. Measuring distances can probe the expansion history of the universe - general relativity connects the expanding spacetime to the mass and energy distribution of the universe. The distance measurements that explain the expansion of the universe (that then point towards the mass-energy content) use measuring Type Ia supernovae distances as “standard candles”.

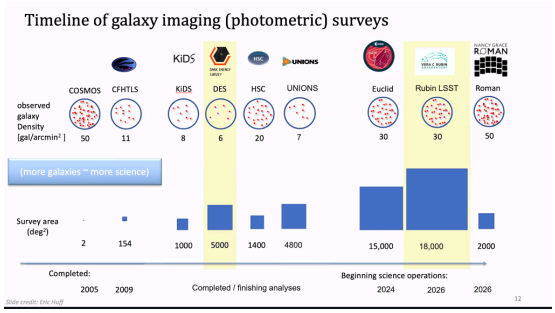
Dark energy can be described as a fluid equation so it can be parameterized for time-dependent behavior. If w = p_{DE} / rho_{DE} which is a constant (-1 for lambda-CDM) but to experiment, view changing w with time.



This plot was analyzed in class earlier to see how we can more effectively and efficiently plot data for both interpretability and accessibility.

The slide shows the timeline for recent expansion history measurements that would prefer models with time-dependent dark energy behavior which is the exciting new results that were spoken about earlier.

In cosmology, people can measure background expansion and structure growth over time. Galaxy surveys provide many methods for experimentation with galaxy clustering and supernovae (Background Expansion), as well as weak lensing and galaxy clusters (Structure Growth). Some others are Time variable AGN, Milky way structure, galactic evolution, strong lensing, solar system objects (TNO and comets), but the talk will not focus on these.



This shows many galaxy imaging surveys, the two highlighted are those she works with. Those with the higher number of galaxies provide more statistical power because of the greater sample size and type of galaxy diversity for example.

To probe structure growth, they study the distribution of matter; the distribution of matter and its evolution over time depends on gravity, properties of dark matter, and expansion (due to dark energy). Using the light that traveled through galaxy structures, it is perturbed due to gravity and you can model the clustering of galaxies by the correlation of signal as a function of the separation on the sky which is the 2 point function. Weak lensing is one percent distortion of galaxy shapes to the line of sight structure. Galaxy clustering is when the galaxy density trances matter density determined by the equation $\text{delta}_{gal} = b * \text{delta}$. Three cross two point analysis is the term that describes the combined study of galaxy lensing, clustering, and weak lensing.

For precision and accurate cosmology, in a bayesian analysis framework, they marginalize over the intrinsic alignments, galaxy bias, photo-z uncertainties, galaxy magnification, and shear calibration, they scale cuts to reduce sensitivity to baryonic feedback, non-linear galaxy bias, and uncertainties in nonlinear P(k), and also do blind analysis to avoid implicit experimenter bias. The coverage of bayesian statistics in this course helps understand real research efforts like this. The DES year 6 galaxy clustering result is a large collaboration filled with early and mid-career scientists.

The pipeline from “pixels to cosmology” has many steps that all need high precision decisions, calculations, and results. In DES they avoid experimenter bias by an equation where blinded data is the measured data plus the model at shifted parameters, subtracted by the model at reference parameters. Posterior predictive distribution tests of internal consistency are used to check the given model and 2pt functions. By comparing the observed data against synthetic data drawn from the model's posterior predictive distribution,they can evaluate the goodness-of-fit and check for tensions or inconsistencies.